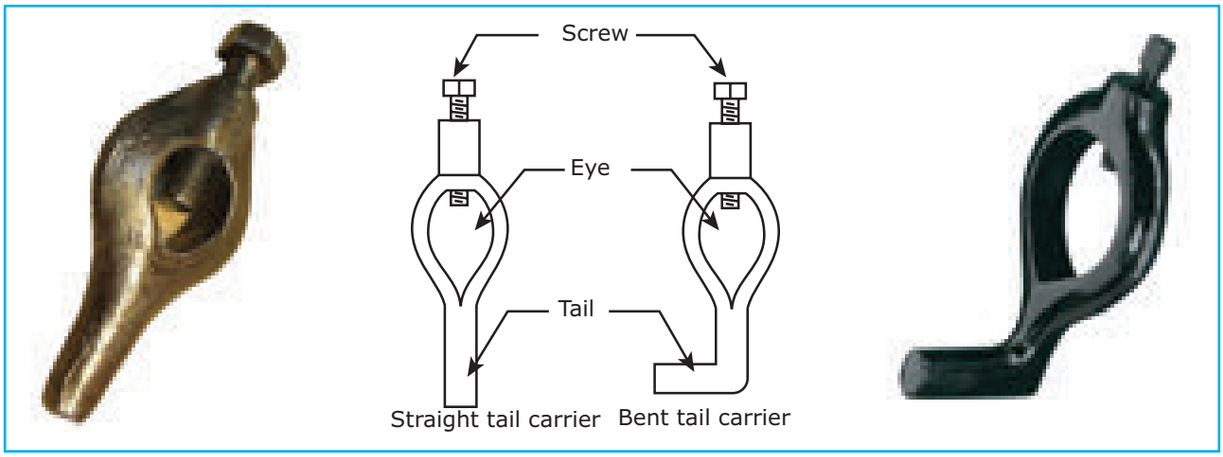


Straight tail carrier

Bent tail carrier

Carrier

When a workpiece is held and machined between centrers, carriers are useful in transmitting the driving force of the spindle to the work by means of driving plates and catch plates. The work is held inside the eye of the carrier and tightened by a screw. Carriers are of two types and they are:

1. Straight tail carrier
2. Bent tail carrier

Straight tail carrier is used to drive the work by means of the pin provided in the driving plate. The tail of he bent tail carrier fits into the slot of the catch plate to drive the work.

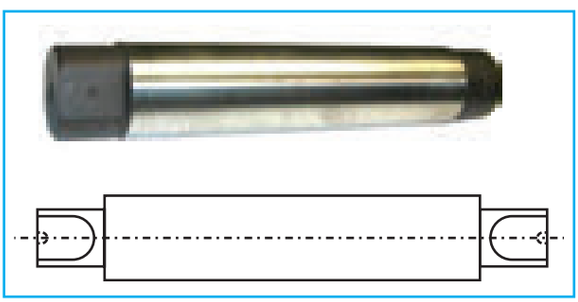
Mandrel

A previously drilled (or) bored workpiece is held on mandrel to be driven in a lathe and machined. There are centre holes provided on both faces of the mandrel. The live centre and the dead centre fit into the centre holes. A carrier is attached at the left side of the mandrel. The mandrel gets the drive either through a catch plate or a driving plate. The workpiece rotates along with the mandrel. There are several types of mandrels and they are:

1. Plain mandrel
2. Step mandrel
3. Gang mandrel
4. Screwed mandrel
5. Collar mandrel
6. Cone mandrel
7. Expansion mandrel

Plain mandrel

The body of the plain mandrel is slightly tapered to provide proper gripping of the workpiece. The taper will be around 1 to 2 mm for a length of 100 mm. It is also known as solid mandrel. This type is mostly commonly used and has wide application.



Plain mandrel

Screwed mandrel

It is threaded at one end and a collar is attached to it. Workpieces having internal threads are screwed on to it against the collar for machining.

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